

OCR Further Mathematics
Pure Core - Series
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
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Question			Answer/Indicative content	Marks	Part marks and guidance		
1			$\frac{1}{(2r-1)(2r+1)} = \frac{A}{2r-1} + \frac{B}{2r+1}$ $\Rightarrow A(2r+1) + B(2r-1) = 1$ $\Rightarrow A - B = 1, A + B = 0$ $\Rightarrow A = \frac{1}{2}, B = -\frac{1}{2}$ $\Rightarrow \sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{1}{2} \left(\left(\frac{1}{1-3} \right) + \left(\frac{1}{3-5} \right) + \dots \right)$ $= \frac{1}{2} \left(1 - \frac{1}{2n+1} \right)$ $\frac{1}{2} \left(1 - \frac{1}{2n+1} \right) \geq 0.49$ $\Rightarrow n \geq 0.98n + 0.49$ $\Rightarrow n \geq \frac{0.49}{0.02} = 24.5$ $\Rightarrow n = 25$	M1 (AO 3.1 a) M1 (AO 1.1) A1 (AO 1.1) M1 (AO 3.1 a) M1 (AO 2.1) A1 (AO 1.1) M1 (AO 3.1 a) A1 (AO 3.2 a) [8]	partial fractions Allow any method to determine A and B Both values Use of differences Deal with subtraction Use of inequality on their formula No marks for a purely numerical solution.		
			Total	8			
2			DR $\sum_{r=1}^{100} (2r+3)^2 = 4 \sum_{r=1}^{100} r^2 + 12 \sum_{r=1}^{100} r + 9 \sum_{r=1}^{100} 1$ $\sum_{r=1}^{100} r^2 = \frac{1}{6} \times 100(100+1)(2 \times 100 + 1)$ $4 \times 338350 + 12 \times \frac{1}{2} \times 100 \times 101 + 900 = 1414900$	B1 (AO3.1 a) M1 (AO1.1 a) A1 (AO1.1) [3]	Expanding and separating Use of formula for $\sum_{r=1}^{100} r^2$		
			Total	3			
3	a		DR	B1(AO1.	Correct	Could be	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$(r+2)(r+1)$ $\frac{A}{r+1} + \frac{B}{r+2}$ $A = 1, B = -1$ $\Sigma = \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} \dots - \frac{1}{n+1} + \frac{1}{n+1}$ $= \frac{1}{2} - \frac{1}{n+2}$	1a) M1(AO1.1) A1(AO1.1) M1(AO2.1) A1(AO1.1)[5]	factorisation of denominator so Correct general form for partial fractions so by correct answer Both Codone omission of $+\frac{1}{4}$ and $-\frac{1}{n+1}$ for M1 only AG. Cancellation must be evident	incorrect if recovered. $\frac{A}{r+1} + \frac{B}{r+2} + C$ and $C = 0$

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Examiner's Comments

Almost all candidates were successful in forming the partial fractions and most recognised the need to use the method of differences. While many clear answers were seen, there were two common errors.

- Not showing sufficient terms to demonstrate cancellation properly.
- Not equating the two sides of the identity.

A few candidates took a perfectly acceptable algebraic approach to get an interim expression, such

$$\text{as } \sum_{r=1}^n \frac{1}{r+1} - \sum_{r=2}^{n+1} \frac{1}{r+1}.$$



Question		Answer/Indicative content	Marks	Part marks and guidance	
	b	DR $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{2}$ since $\frac{1}{n+2} \rightarrow 0$ as $n \rightarrow \infty$	B1(AO2.2a) B1(2.4) [2]	$O \lim_{n \rightarrow \infty} \left(\frac{1}{n} - \frac{1}{n+2} \right) = \frac{1}{2} - 0 = \frac{1}{2}$ r Indication that $1/(n+2)$ is close to zero (accept “small”) when n is large <u>Examiner’s Comments</u> Although candidates generally scored well on this question, a lot of conceptual misunderstandings were in evidence (which would result in greater problems in Question 3). Many candidates seemed confused about the notion of limit and statements using infinity as a number were common, as were statements such as $\sum_{n=1}^{\infty} \dots \rightarrow \frac{1}{2}.$	
		Total	7		

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Question		Answer/Indicative content	Marks	Part marks and guidance		
4	a	DR $(r+2)(r-1)$ $\frac{A}{r-1} + \frac{B}{r+2}$ $A = 1, B = 1$ $\therefore \sum_{r=5}^n \frac{3}{r^2+r-2} = \sum_{r=5}^n \frac{1}{r-1} - \sum_{r=5}^n \frac{1}{r+2}$ $= \sum_{r=5}^n \frac{1}{r-1} - \sum_{r=8}^{n+3} \frac{1}{r-1}$ $= \sum_{r=5}^7 \frac{1}{r-1} - \sum_{r=n+1}^{n+3} \frac{1}{r-1}$ Alternative method for last 2 marks: $= \frac{1}{4} - \frac{1}{7} - \left(\frac{1}{n-1} - \frac{1}{n+2} \right)$ $= \frac{1}{4} - \frac{1}{7} - \frac{1}{n-1} + \frac{1}{n+2}$	B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 A1 [5]	Correct factorisation of denominator so correct form for partial fractions Using partial fractions, separating into two sums, re-indexing so that the summands have identical form and cancelling central terms.	Might see start and end terms explicitly. eg $\sum_{r=5}^n \frac{1}{r-1} - \sum_{r=8}^{n+3} \frac{1}{r-1}$ Might see formal substitution of index. eg Let $R = r + 3 \Rightarrow r + 2 = R - 1$ $\therefore \sum_{r=5}^n \frac{1}{r-1} - \sum_{r=5}^n \frac{1}{r+2}$ $= \sum_{r=5}^n \frac{1}{r-1} - \sum_{R=8}^{n+3} \frac{1}{R-1}$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$= \frac{1}{4} + \frac{1}{5} + \frac{1}{6} - \frac{1}{n} - \frac{1}{n+1} - \frac{1}{n+2}$ $= \frac{37}{60} - \frac{1}{n} - \frac{1}{n+1} - \frac{1}{n+2}$		At least these terms	M1 can be ft from any A, B having opposite signs For M1, condone omission of $\frac{1}{7}$ or $-\frac{1}{n-1}$ Requires joined argument Must have either clear diagonal cancellations with one explicit cancellation or less clear cancellation with numerical and algebraic cancellation shown or described

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Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		$= \frac{37}{60}$ or awrt 0.617	B1 (AO2.2a) [1]	<u>Examiner's Comments</u> The mark scheme gives an algebraic demonstration of method of differences as the most efficient approach. Often, historically, teaching and textbooks have focused on the alternative approach on the mark scheme – namely one which writes down several numerical and algebraic terms from the sum and shows subsequent evidence of cancellation through crossing out. This alternative approach will continue to receive credit, but centres may wish to consider teaching the algebraic approach – certainly stronger candidates will enjoy the mathematical precision, and the completeness of the method will reduce any uncertainty as to just how many terms are required to convincingly demonstrate the alternative approach.	
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
5			Basis case: $1^3 + 2^3 + 3^3 = 36 = 4 \times 9$ Consider the sum $f(r) = r^3 + (r + 1)^3 + (r + 2)^3$ Assume that $f(r) = 9k$ for some $k \in \mathbb{Z}$ Then $f(r + 1) = f(r) + (r + 3)^3 - r^3$ $= f(r) + r^3 + 3r^2 \times 3 + 3r \times 9 + 27 - r^3$ $= f(r) + 9r^2 + 27r + 27$ $= 9k + 9(r^2 + 3r + 3) = 9k'$ for some $k' \in \mathbb{Z}$ So if true for r then true also for $r + 1$ But it is true for $r = 1$ so is true for all integers r	B1 (AO2.1) M1 (AO2.1) M1 (AO2.1) A1 (AO2.2a) A1 (AO2.5) [5]	Basis case - either 4×9 or "36 is divisible by 9" given explicitly. For getting started (Sight of $f(r)$ is not necessary) For finding $f(r + 1)$ Each term as a multiple of 9 Conclusion dependant on all other marks being earned <u>Examiner's Comments</u> Many candidates completed this question successfully. However, a significant number of candidates dropped marks with the formal statements required for an inductive proof, either in the initial base case or in the final conclusion.	
			Total	5		

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candidates were able to use the formulae for the sums of cubes, squares and integers and to make good progress to achieve the correct response. These



Question			Answer/Indicative content	Marks	Part marks and guidance	
					The remainder required double the amount of work by multiplying out and then factorising with some dropped marks through careless mistakes.	
			Total	3		
7			$\sum_{r=1}^{10} r(3r-2) = \sum_{r=1}^{10} (3r^2 - 2r) = 3 \sum_{r=1}^{10} r^2 - 2 \sum_{r=1}^{10} r$ $= 3 \left(\frac{1}{6} 10 \cdot 11 \cdot 21 \right) - 2 \left(\frac{1}{2} 10 \cdot 11 \right)$ $= 55(21 - 2) = 55 \times 19$ $= 1045$ Alternative Q3 leaving substitution to the end $\sum_{r=1}^{10} r(3r-2) = \sum_{r=1}^{10} (3r^2 - 2r) = 3 \sum_{r=1}^{10} r^2 - 2 \sum_{r=1}^{10} r$ $= \frac{3}{6} n(n+1)(2n+1) - \frac{2}{2} n(n+1)$ $= \frac{1}{2} n(n+1)(2n+1-2) = \frac{1}{2} n(n+1)(2n-1)$ $n=10 \Rightarrow \frac{1}{2} 10 \cdot 11 \cdot 19$ $= 1045$	M1(AO1.1) M1(AO2.1) A1(AO1.1) [3]	Separate soi oe 1155 - 110 Use both formulae with $n = 10$ AG <u>Examiner's Comments</u> The vast majority of candidates understood the meaning of the command words 'show that' and gave enough detail in their answer to show that ten terms had not been added together.	
			Total	3		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
8		a	$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = u$ $\Rightarrow e^x(1 - u) = e^{-x}(1 + u)$ $\Rightarrow e^{2x} = \left(\frac{1+u}{1-u}\right) \Rightarrow x = \frac{1}{2} \ln \left(\frac{1+u}{1-u}\right)$	M1 (AO 3.1a) A1 (AO 2.1) M1 (AO 3.1a) A1 (AO 1.1) [4]	Use of exponentials Attempt to find e^{2x}	
		b	$4 \tanh^2 x + \tanh x - 3 = 0$ $\Rightarrow (4u-3)(u+1) = 0 \Rightarrow u = \frac{3}{4}, (-1)$ $\Rightarrow x = \frac{1}{2} \ln \left(\frac{1+\frac{3}{4}}{1-\frac{3}{4}} \right) = \frac{1}{2} \ln 7 \quad \text{oe}$ i.e. $a = \frac{1}{2}, b = 7 \quad \text{oe}$	M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) [4]	Solve quadratic Substitute into the standard form	
		c	E.g. Because $-1 < \tanh x < 1$	B1 (AO 2.4) [1]	e.g. One root has been rejected as outside range.	
Total				9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
9		a	$\frac{1}{r(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+2}$ $\Rightarrow A(r+1)(r+2) + Br(r+2) + Cr(r+1) = 1$ $\Rightarrow A = \frac{1}{2}, B = -1, C = \frac{1}{2}$ $\Rightarrow \frac{1}{r(r+1)(r+2)} = \frac{1}{2} \left(\frac{1}{r} - \frac{2}{r+1} + \frac{1}{r+2} \right)$ $\sum_{r=1}^n \frac{1}{r(r+1)(r+2)} =$ $= \frac{1}{2} \left[\left(\frac{1}{1} - \frac{2}{2} + \frac{1}{3} \right) + \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) + \dots \right]$ $= \frac{1}{2} \left[\left(\frac{1}{n} - \frac{2}{n+1} + \frac{1}{n+2} \right) \right]$ $= \frac{1}{2} \left[\left(\frac{1}{1} - \frac{2}{2} + \frac{1}{2} \right) + \left(-\frac{1}{n+1} + \frac{1}{n+2} \right) \right]$ $= \frac{1}{4} - \frac{1}{2(n+1)} + \frac{1}{2(n+2)}$	M1 (AO 3.1a) M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 1.1) M1 (AO 2.1) A1 (AO 1.1) [6]	Attempt to find partial fractions Attempt to find constants by comparing 3 coefficients or substituting 3 values for r Expand sum using their partial fractions Cancel terms Answer in the form $\frac{1}{4} - \frac{1}{2}f(n)$	
		b	$\frac{1}{4}$	B1 (AO 2.2a) [1]		
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
10			$\frac{z+7i}{z-24} = \frac{x+iy+7i}{x-24+iy} \times \frac{x-24-iy}{x-24-iy}$ $\operatorname{Im} \frac{z+7i}{z-24} = \frac{-xy + (y+7)(x-24)}{(x-24)^2 + y^2} = \frac{1}{4}$ $28x - 96y - 672 = x^2 - 48x + 576 + y^2$ $0 = (x-38)^2 - 1444 + (y+48)^2 - 2304 + 1248$ $(x-38)^2 + (y+48)^2 = 2500$ So the shape of C is a circle... ...centre $38 - 48i$, radius 50	M1 (AO 3.1a) M1 (AO 2.1) M1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.2a) E1 (AO 3.2a) E1 (AO 3.2a) [7]	Substituting $z = x + iy$ into $\frac{z+7i}{z-24}$ and multiplying top and bottom by conjugate of bottom Multiplying out (ignore errors in real part) and equating imaginary part to $\frac{1}{4}$ (without i unless later cancelled or recovered) Multiplying out to get horizontal equation with no xy term and no double brackets Completing both squares with half signed coefficients of x and y Or (38, -48)	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance		
11		a	$\frac{1}{(2r-1)(2r+1)} = \frac{A}{(2r-1)} + \frac{B}{(2r+1)}$ $= \frac{1}{2} \left(\frac{1}{(2r-1)} - \frac{1}{(2r+1)} \right)$ $= \frac{1}{2} \left(\frac{1}{1} - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \dots + \frac{1}{2n-1} - \frac{1}{2n+1} \right)$ $= \frac{1}{2} \left(\frac{1}{1} - \frac{1}{2n+1} \right) = \frac{n}{2n+1}$	M1 (AO3.1a) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1)	Partial fractions Use method of differences		
		b	e.g. $\frac{n}{2n+1} = \frac{1}{2} \times \frac{2n+1-1}{2n+1}$ $= \frac{1}{2} \times \left(1 - \frac{1}{2n+1} \right) \rightarrow \frac{1}{2} \text{ as } n \rightarrow \infty$	B1ft (AO2.2a) [1]	FT from their answer providing it is of the form $\frac{an+b}{cn+d}$		
			Total	5			



Question			Answer/Indicative content	Marks	Part marks and guidance		
12		a	$\sum_{r=1}^n ((r+1)^5 - r^5) = (n+1)^5 - 1$ $(r+1)^5 - r^5 = 5r^4 + 10r^3 + 10r^2 + 5r + 1$ Use of $\sum_{r=1}^n 1 = n$ and $\sum_{r=1}^n r = \frac{n(n+1)}{2}$ $(n+1)^5 - 1 = 5 \sum_{r=1}^n r^4 + \frac{5n^2(n+1)^2}{2} + \frac{5n(n+1)(2n+1)}{3} + \frac{5n(n+1)}{2} + n$ $30 \sum_{r=1}^n r^4 = 6n(n^4 + 5n^3 + 10n^2 + 10n + 4) - n(n+1)(15n(n+1) + 15 + 10(2n+1))$ Further factorisation leading to $\sum_{r=1}^n r^4 = \frac{1}{30} n(n+1)(2n+1)(3n^2 + 3n - 1)$	B1 (AO 1.1a) M1 (AO 3.1a) B1 (AO 1.1) M1 (AO 3.1a) M1 (AO 1.1) A1 (AO 2.1) [6]	Binomial expansion of $(r+1)^5$ with r^5 cancelling Correct cancellation of sum down to two terms and substitution Expansion and rearrangement leading to (eg) n as being a common factor No gaps in argument for A1 AG	or attempt at comparable expansions of $\sum_{r=1}^n r^4$ expression and AG. or both expansions correctly and completely demonstrated as equal WWW.	
		b	$\sum_{r=1}^{80} r^4 - \sum_{r=1}^{49} r^4$ $(676\ 010\ 664 - 59\ 416\ 665)$ $\Rightarrow 616\ 593\ 999$	M1 (AO 2.2a) A1 (AO 1.1) [2]	soi		
			Total	8			



Question			Answer/Indicative content	Marks	Part marks and guidance		
13			DR $\frac{1}{6} \times 300 \times (300 + 1)(2 \times 300 + 1)$ $\frac{1}{6} \times 120 \times (120 + 1)(2 \times 120 + 1)$ $(= 9045050 - 583220) = 8461830$	M1 (AO 1.1a) M1 (AO 2.1) A1 (AO 2.2a) [3]	Any correct numerical use of formula Must be 120, not 121 (unless corrected by 121 ² later).	Implied by 9045050 seen Implied by 583220 seen	
			Total	3			



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Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	$\frac{2r+5-(2r-1)}{(2r-1)(2r+5)}$ or $\frac{2r+5-2r+1}{(2r-1)(2r+5)}$ must be seen	B1 [1]	Derive given result correctly (or use complete partial fractions method) N.B. 1 st expression could be 2 separate fractions Examiner's Comments The majority of candidates showed sufficient working to justify the derivation of the given answer. A numerator of $2r+5-2r-1$ was the most common error.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	<i>Either</i> 0.629 <i>Or</i> $\sum_{r=2}^{30} \frac{1}{2r-1} = 1.68237.. \quad \sum_{r=2}^{30} \frac{1}{2r+5} = 1.05383..$ 0.629	M1* A1 A1 DM1 A1 [5] M1 A1 A1 M1 A1	Express at least 2 terms as differences using (i), could start at $r = 1$ Obtain $\frac{1}{3}, \frac{1}{5}, \frac{1}{7}$, obtain $-\frac{1}{61}, -\frac{1}{63}, -\frac{1}{65}$ these may be unsimplified Show correct cancelling (and subtraction of 1st term if appropriate) Obtain correct answer Attempt to find sum of 2 separate series, could start at $r = 1$ Obtain correct answers, must be at least 4 decimal places Subtract their values Obtain correct answer Examiner's Comments Those who started the difference method at $r = 2$ generally found the correct cancelling and then the correct value to 3 decimal places. Those who started at $r = 1$, usually made an error in the cancelling or did not deal with the subtraction of the first term from their expression properly.	
		iii	71/105	B1 [1]	Obtain correct answer Examiner's Comments A sensible attempt at (ii) usually meant a correct answer was given for this part.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	7		
15			$\frac{1}{6}n(n+1)(2n+1) - \frac{1}{2}n(n+1) - 8n$ $\frac{1}{6}n(2n^2 - 50)$ $\frac{1}{3}n(n-5)(n+5)$	M1 A1 M1 A1 A1 [5]	Use at least 2 correct standard results, must have 3 terms Obtain correct unsimplified answer Attempt to factorise, must get at least a factor of n and simplify Obtain correct answer Obtain correct final answer	Examiner's Comments Most candidates used the correct standard results, the most common error was to give the last term as 8, rather than $8n$. This inevitably meant that their result could not be factorised. A significant number of those who found a correct expression, did not see the factor of 2, or did not factorise $(n^2 - 25)$.
			Total	5		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
16		a	$\left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \dots$ $+ \left(\frac{1}{n-1} - \frac{1}{n+1}\right) + \left(\frac{1}{n} - \frac{1}{n+2}\right)$ Cancelling of inner terms $\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2}$	M1(AO2.1) M1(AO1.1) A1(AO1.1) [3]	At least first and last terms in full		
		b	$\frac{1}{n+1} \rightarrow 0$ and $\frac{1}{n+2} \rightarrow 0$, hence sum $\rightarrow \frac{3}{2}$	E1(AO2.4) [1]		Justification required E0 for unsupported answer	
			Total	4			
17			$\sum_{r=1}^n r^2 + 6 \sum_{r=1}^n r + \sum_{r=1}^n 5$ $= \frac{1}{6}n(n+1)(2n+1) + 6 \frac{n(n+1)}{2} + 5n$ $\frac{1}{6}n(n+7)(2n+7) \text{ oe}$	M1(AO1.1a) M1(AO1.2) A1(AO1.1) A1(AO1.1) [4]	At least two of three elements correct Substitution of at least one correct standard formula Correct unsimplified answer	For full marks all algebra must be correct	
			Total	4			



Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i		M1	Difference of sum to $2n$ and $n - 1$	
		i		A1	Correct unsimplified answer	$\frac{1}{2}(2n)^2(2n+1)^2 - \frac{1}{2}(n-1)^2n^2$
		i		M1	Sensible attempt to factorise, at least factor n^2	
		i		A1	Obtain given answer no errors seen	
					<u>Examiner's Comments</u>	
					Many candidates did not realise that the sum to $(n - 1)$, not n , needed to be subtracted and then just assumed that they could obtain the given answer. If they had been more careful, the mistake could have been identified, and possibly rectified.	
		ii		M1	Difference of (i) and another standard result	
		ii		M1	Difference of $\sum_{1}^{2n} r - \sum_{1}^k r$ for $k = n - 1$ or n	$\frac{1}{2}(2n)(2n+1) - \frac{1}{2}n(n-1)$
		ii		A1	Obtain complete unsimplified expression	(i) - $2 \times$ above
		ii		M1	Sensible attempt to factorise, at least factor $n(n + 1)$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{3}{4}n(n+1)^2(5n-4)$	A1	Obtain correct answer <u>Examiner's Comments</u> Having obtained an unsimplified expression for the required sum, many candidates expanded all terms, before trying to factorise and usually made an error in the process. Some did not use the answer from (i), which meant that the simplification and factorisation became more difficult.	$(n+1)(n+1)$ is OK for $(n+1)^2$
			Total	9		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
19			$\frac{1}{6}n(n+1)(2n+1)-n$	M1*	Attempt to expand $(r-1)(r+1)$	
				DM1	Use standard result for Σr^2	
				A1	Obtain correct unsimplified answer	
			$\frac{1}{6}n(2n+5)(n-1)$	DM1	Attempt to factorise	
				A2	Obtain completely correct answer Allow A1 if one bracket still contains a common factor Examiner's Comments Completely correct solutions were given by a good number of candidates. Most attempted the difference of two terms, with $\Sigma 1$ being given as 1 rather than n the most common error. A significant minority expanded the correct expression to obtain a cubic, before attempting to factorise, rather than using a common factor of n or $\frac{1}{6}n$ as the first stage of factorisation.	
Total			6			

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Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i		M1	Obtain correct numerator from addition or partial fractions	
		i		A1	Obtain given answer correctly	
					Examiner's Comments	
					The majority of candidates showed sufficient working to justify the given answer.	
		ii		M1	Express at least three relevant terms using (i)	
		ii		A1	1 st three terms correct	
		ii	$\frac{n}{(n+1)(n+2)}$	A1	Last two terms correct	
		ii		M1	Show correct cancelling	
		ii		A1	Obtain given answer correctly	
					Examiner's Comments	
					Candidates who displayed the terms in columns usually showed the correct cancelling process. Those who listed terms in rows often had difficulty in seeing the correct way to cancel.	
		iii	$-\frac{1}{6}$	M1	Sum 1 to ∞ – 1 st term or start process at $r = 2$	
		iii		A1	Obtain correct answer	
					Examiner's Comments	
					Many candidates thought that the sum from 1 to ∞ was 1, rather than the correct value of 0. Some did not see that the sum from 2 to ∞ was required.	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	$\frac{2}{3}, \frac{2}{5}, \frac{2}{7}$	B1	B1 $\times$ 3, Obtain 3 correct values	
		i		B1		
		i		B1	Justify given answer	
					Examiner's Comments	
					Most found the first two answers correctly, but too frequently no working was shown to justify the value for u_4 .	
		ii	$\frac{2}{2n-1}$	M1	Fraction, in terms of n , with correct numerator or denominator	
		ii		A1	Obtain correct answer a.e.f.	
					Examiner's Comments	
					A correct expression, in terms of n , was usually seen, but a minority gave an expression in terms of u_n .	
		iii	$\frac{2}{2(n+1)-1}$	B1ft	Verify result true when $n = 1$, for their (ii), or $n = 2, 3$ or 4	
		iii		M1	Expression for u_{n+1} using recurrence relation in terms of n using their (ii)	
		iii		A1	Correct unsimplified answer	
		iii		A1	Correct answer in correct form	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		B1	Specific statement of induction conclusion, previous 4 marks must be earned, $n = 1$ must be verified Examiner's Comments Sufficient working to demonstrate that the result was true for $n = 1$ was generally seen, as was the working to derive the correct expression for u_{k+1} from the recurrence relation and their part (ii). However, candidates should be made aware that the work required in an induction question is mainly of the "given answer" type, so sufficient justification is essential. Many candidates did not give a satisfactory conclusion to the induction process.	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	Construct rectangles from $x = 1$ of height y to n or ∞	M1	Or sum from 1 to n so from diagram	Or to $n - 1$
		i	Sum of areas of rectangles $= \frac{1}{1} + \frac{1}{2} + \dots = \sum_1^{\infty} \frac{1}{x}$			
		i	This area is bigger than the area under the curve	A1	For sum and inequality stated or implied by diagram	Condone comparison of areas for 1 to n and 1 to $n - 1$
		i	$A = \int_1^{\infty} \frac{1}{x} dx = [\ln x]_1^{\infty} = \infty$	B1		
		i	Since Sum $> A$, the sum is infinite.	A1	Integral from 1 to $n + 1$ or ∞ Conclusion	If $n - 1$ above then this integral to n Examiner's Comments Those candidates who produced concise, efficient solutions almost invariably started with a well drawn diagram where the heights of the rectangles were clearly equivalent to equally spaced ordinates. Many candidates produced unsatisfactory diagrams with seemingly random rectangles placed above the curve and, although they recognised that their summation was greater than an integral they made heavy weather of producing a fully correct solution. Whilst many candidates who did not draw a diagram were also able to get a full solution, these tended to start off by looking at the area of n rectangles compared to an integral between 1 and $n + 1$ (or

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Question			Answer/Indicative content	Marks	Part marks and guidance	
						even just n) before moving on to look at the infinite case thereby making their solution lengthier. Some candidates started off by comparing a specific number of rectangles with a corresponding integration and continued to work numerically. Few of these went on to successfully extend to the infinite case. Many other candidates drew rectangles below the curve and argued that since their integration gave an infinite value then an infinite series of rectangles that were being used as an approximation must also be infinite, some of these actually went on to do part(ii) correctly and yet did not realise that this highlighted the fallacy in their reasoning. Other candidates drew two series of rectangles, one above and one below the curve, clearly believing that it was necessary to provide upper and lower bounds. Nearly all candidates were able to carry out the integration correctly.
		ii	Construct rectangles from $x = 2$ to the left of height y to n or ∞ Sum of areas of rectangles $= \frac{1}{2^2} + \frac{1}{3^2} + \dots = \sum_2^{\infty} \frac{1}{x^2}$	M1	Rectangles must be under curve	May include the rectangle from $x = 1$ to the left



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	This area is less than the area under the curve $A = \int_1^{\infty} \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_1^{\infty} = 0 - -1 = 1$	A1	For sum and inequality stated or implied by diagram	Examiner's Comments As in part (i) those candidates who started with a well drawn diagram tended to get a complete solution with very little extra work required. Adding in the extra 'first' rectangle and shading it was a commonly used effective strategy and for most successful candidates the '+1' required appeared early in their working rather than towards the end. There were some candidates, however, who would add 1 to the result of their integration but without any apparent understanding of why. It is possible that they had realised that $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$ was clearly > 1 and probably < 2 so it was a good thing to do or it may have been that they had a recollection of using this type of method in other examples.
		ii	Since $\text{Sum} < A$, $f(x) = \frac{x(x-1)}{(x+1)(x^2+1)}$	B1	Area under curve	
		ii	$\Rightarrow \sum_1^{\infty} \frac{1}{x^2} < 1 + 1$	M1		
		ii	Upper limit = 2	A1	For adding 1 to both sides, may appear earlier	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
						Many candidates who had drawn rectangles under the curve in part(i) reversed their error in part(ii) and drew them above this time. As in part(i) two series of rectangles were often seen. A worryingly large number of candidates integrated incorrectly to get $\left[-2x^{-3}\right]_1^{\infty}$; by coincidence this gives them the answer of 2 immediately which may have convinced them they were correct at that point. Others carried out their integration with 2 as the lower limit (easily confused with the 2 for their first rectangle) and obtained an answer of 1/2 which then becomes 3/2 once they add on their extra rectangle (which they all did since they weren't going to use 2 without appreciating the need for an extra rectangle).
Total				9		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	Heights of rectangles = $\left(\frac{1}{2}\right)^3, \left(\frac{1}{3}\right)^3, \left(\frac{1}{4}\right)^3, \dots, \left(\frac{1}{n}\right)^3$ (Width of rectangles = 1)	M1	Heights, with at most one extra and / or one omitted	
		i	$\Rightarrow$ Sum of areas = $\left(\frac{1}{2}\right)^3 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{4}\right)^3 + \dots + \left(\frac{1}{n}\right)^3$ or $\sum_{r=2}^n \left(\frac{1}{r}\right)^3$ or $\sum_{r=1}^n \left(\frac{1}{r}\right)^3 - 1$	A1	isw	No limits M0
		ii	Area = $\int_1^{\infty} \frac{1}{x^3} dx = \left[-\frac{1}{2x^2} \right]_1^{\infty} = \frac{1}{2}$	M1	Integrate correct function:	Or with upper limit of n
		ii		A1	seen by x^2 in denominator	
		ii	Since sum of areas of rectangles approximates, but is less than, the area under the curve	M1	www	
		ii			Compare <i>their</i> answer to (i) (taken to ∞) with <i>their</i> integral dep on 1st M	
		ii	$\left(\frac{1}{2}\right)^3 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{4}\right)^3 + \dots = \sum_{r=2}^{\infty} \left(\frac{1}{r^3}\right) < \frac{1}{2}$	M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow \sum_{r=1}^{\infty} \left(\frac{1}{r^3} \right) < \frac{1}{2} + 1 = \frac{3}{2}$	A1	Dealing with 1 Dep on previous 2 Ms Examiner's Comments Most could write down the sum of areas in part (i), some expressing it using the summation notation which was commendable. However, in part (ii) there was considerable confusion. Some started talking about an "upper bound" (which of course is nothing to do with the question). The most common error however was to see the extra 1 as an extra rectangle. It was necessary then to change the limits of the integral. The result of this differentiation cannot be evaluated but by this stage all that could be done was to fudge it. Those that used the summation notation with upper and lower values usually realised that what they had which was equal to $\frac{1}{2}$ was not the summation of the question and the first term (which was 1) had to be added.	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
24			$\frac{1}{2}n(n+1)(2n+1) - n(n+1) - n$ $\frac{1}{2}n(2n+3)(n-1) \text{ or } n(n+\frac{3}{2})(n-1)$	M1* A1 A1 DM1 A1	Expand and attempt to use standard series, at least one used correctly Any two terms correct, may be unsimplified All terms correct Attempt to find 3 factors Obtain correct answer Examiner's Comments Most candidates expanded and attempted to use the correct standard results, although some used only the formula for Σr and then expanded. Many made algebraic slips, especially in the middle term, which lead to an expression that could not be factorised and many made an error when trying to take out a factor of $1/2$.	
			Total	5		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
25		i	$\frac{2r+3-(2r+1)}{(2r+1)(2r+3)}$ or $\frac{2r+3-2r-1}{(2r+1)(2r+3)} = \frac{2}{(2r+1)(2r+3)}$	B1	Establish given result correctly, might be a "0 = 0" type verification, or partial fractions	
					Examiner's Comments	
					Most showed sufficient working to justify the given result. A numerator of $2r+3-2r+1$ was the most common error.	
		ii		M1	Express terms as differences using (i)	
		ii		M1	Attempt this for at least 1 st two and last term	
		ii	$1/3 - 1/5$ and $1/5 - 1/7$	A1	First two terms correct, do not penalise missing factor of 2	
		ii	$1/(2n+1) - 1/(2n+3)$	A1	Last term correct, do not penalise missing factor of 2, allow in terms of e.g. r or	
		ii		M1	k	
		ii	$\frac{n}{3(2n+3)}$	A1	Show correct cancelling Obtain a correct single fraction (denominator could be expanded) must be in terms of n. N.B. Be on look out for f(r) - f(r+1) approach	
					Examiner's Comments	
					The method of differences was understood by most candidates. Many did not give the answer as a single fraction and a good proportion failed to include the required factor of 1/2 in their final answer.	
		iii	Either $\frac{1}{6} - \frac{n-1}{3(2n+1)}$	M1	Use sum to ∞ - sum to $n-1$ or n	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		A1	Obtain correct unsimplified expression, do not penalise missing factor of 2	
		iii	$\frac{1}{2(2n+1)}$	A1	Obtain a correct single fraction (denominator could be expanded)	
		iii	Or	M1	Start differences at $r = n$	
		iii	$\frac{1}{(2n+1)}$	A1	Obtain correct remaining term	
		iii	$\frac{1}{2(2n+1)}$	A1	Obtain a correct single fraction (denominator could be expanded)	
					Examiner's Comments This part proved quite testing. Most candidates tried to find the sum to infinity but many got 0 or 1 for this and many subtracted the sum to n rather than the sum to $n-1$. Again the factor of $1/2$ was often omitted.	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
26			$\frac{1}{2}n(n+1)(2n+1) - 5n$ $\frac{n(n-\frac{1}{2})(n+3)}{2}$ $\frac{1}{2}n(2n-3)(n+3)$ or	M1* A1 DM1 A1	Express as difference using standard result for $\sum r^2$ Correct unsimplified expression Obtain at least factor of n Obtain correct answer, only these versions Examiner's Comments Few candidates quoted an incorrect expression for $\sum r^2$ and most subtracted $5n$ rather than 5. Most made a sensible attempt to factorise, but a significant number stopped at $\frac{1}{2}n(2n^2+3n-9)$. Those who expanded to obtain a cubic expression usually then made an algebraic error in their attempt to factorise.	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i		M1	Use correct common denominator, numerator must be quadratic	
		i		A1	Obtain given result Examiner's Comments This part was well-done by most candidates, with only minor algebraic slips occurring.	
		ii		M1	Express terms as differences using (i)	
		ii		M1	Attempt this for at least first 3 terms	
		ii		A1	First 3 terms all correct	
		ii		A1	Last 2 terms correct	
		ii		M1	Show terms cancelling	
		ii	$\frac{7}{2} - \frac{3}{n} - \frac{1}{n+1}$	A1	Obtain correct answer, must be in terms of n Examiner's Comments Most candidates used the method of differences correctly, but some started at $r = 1$, missing that this gave an infinite value for the first term and some finished at $r = n + 1$.	Need not be tidied up
		iii	$\frac{5}{4}$	M1	Attempt to start summation at correct term	Could be



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		A1	Obtain correct answer from correct working Examiner's Comments This part was less well-done. Many simply let $n \rightarrow \infty$ in the answer to (ii) or subtracted too many terms from the sum to infinity. Those who started the method of differences at the correct term found the correct answer quite easily.	$\Sigma_{\infty} 2 - \Sigma_2^3$
			Total	10		
28			$4 \times \frac{1}{4} n^2 (n+1)^2 - 3 \times \frac{1}{6} n(n+1)(2n+1) + \frac{1}{2} n(n+1)$ $n^3 (n+1)$	M1 A1 A1 M1 A2	Express as sum of three series Obtain 2 correct (unsimplified) terms Obtain correct 3 rd (unsimplified) term Attempt to factorise, at least factor of n Obtain correct answer, A1 if not fully factorised Examiner's Comments The three standard series were given correctly by most candidates. A fair proportion then expanded each term before attempting to factorise and any sign error meant that their expression would not factorise easily. Even those who used a common factor of $n(n+1)$ still made sign or arithmetic slips, again often obtaining an expression that could not be factorised.	
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i		M1	Use correct denominator or partial fractions	
		i		A1	Obtain given answer convincingly	
					Examiner's Comments	
					This part was answered correctly by the majority of candidates. Marks were lost by those who did not show sufficient correct working to justify the numerator in the answer given in the question; $3r + 2 - 3r - 1 = 3$ was frequently seen.	
		ii		M1	Express at least 1 st two and last term using (i)	
		ii		A1	All terms correct	
		ii		M1	Show correct terms cancelling	
		ii	$\frac{1}{2} - \frac{1}{6n+2}$	A1	Obtain correct unsimplified answer	
		ii		M1	Include $\frac{1}{3}$ and combine their sum as a single fraction	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	Obtain given answer Examiner's Comments Some candidates tried to use the standard series results in the denominators. The majority used the method of differences, but many stopped at the n th term rather than the $2n$ th term, and a good proportion did not deal with the extra factor of 3, even though the answer is given in the question.	
			Total	8		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i		M1	Combine with a correct denominator	
		i		A1	Obtain given answer correctly	
					<u>Examiner's Comments</u>	
					Most candidates showed sufficient working to derive the given answer.	
		ii		M1	Express as differences using (i)	
		ii		M1	Attempt this for at least first 3 terms	
		ii		A1	First 3 terms all correct	
		ii		A1	Last 2 terms all correct	
		ii		M1	Show correct cancelling	
		ii	$1 + \frac{1}{4} - \frac{1}{(n+1)^2} - \frac{1}{(n+2)^2}$	A1	Obtain correct answer i.s.w.	Final answer must be in terms of n
					<u>Examiner's Comments</u>	
					A significant proportion of candidates did not set out the method of differences in a sufficiently detailed way, so that incorrect cancelling of terms occurred.	
		iii	$\frac{61}{900}$	M1	Start differences at $n = 5$ or $S_{\infty} - S_4$	$1 + \frac{1}{4} - (1 - \frac{1}{4} - \frac{1}{5^2} - \frac{1}{6^2})$
		iii		A1	Obtain correct answer, with no errors seen	
					<u>Examiner's Comments</u>	
					Candidates either started the method of differences at the correct term, or subtracted the correct value from the sum to infinity.	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	10	



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